Statistiques avancées 3A

CentraleSupélec

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Outline

1. Introduction
2. Theoretical study of the Lasso in the linear regression model
3. Theoretical study of the Lasso in a non-linear regression model
4. Generalized Linear Model and the Lasso

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Introduction

Generalized Linear Models (GLM) [Nelder and Wedderburn,1972 [3]]:

- include gaussian linear, logit and log-linear models
- express the expectation of the variable of interest Y as a function of a linear combination of the covariates

modèle linéaire : $\mathbb{E}[Y|X=x] = x^T\beta$

modèle linéaire généralisé :

$g(\mathbb{E}[Y|X=x]) = x^T\beta$

The exponential family

Definition

We say that the random variable Y belongs to exponential family if its density can be written as follows:

$$f_\theta(y) = \exp\left(\eta(\theta)\, T(y) - A(\theta) + B(y)\right),$$

where η, T, A, B are deterministic functions, θ is the unknown parameter.

Examples:

1) Normal distribution $\mathcal{N}(\mu, 1)$:

$$f_\mu(y) = \text{à compléter}$$

2) Bernoulli distribution $\mathcal{B}(p)$:

$$f_p(y) = \text{à compléter}$$

$Y \sim \mathcal{N}(\mu, 1)$ $\quad f_\theta(y) = \exp(\eta(\theta)\, T(y) - A(\theta) + B(y))$

$$f_\mu(y) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{(y-\mu)^2}{2}\right)$$

$$= \exp\left(y\mu - \frac{1}{2}\mu^2 - \log(\sqrt{2\pi}) - \frac{y^2}{2}\right)$$

$\theta = \mu$, $\quad \eta(\theta) = \theta$

$T(y) = y$, $\quad A(\theta) = \frac{\theta^2}{2}$,

$B(y) = -\frac{y^2}{2} - \log\sqrt{2\pi}$

2) $Y \sim \mathcal{B}(p)$

$$f_p(y) = p^y (1-p)^{1-y} = (1-p) \exp\left(y \log \frac{p}{1-p} \right)$$

$$\theta = p \quad , \quad \eta(\theta) = \log \frac{\theta}{1-\theta} \quad , \quad T(y) = y$$

$$A(\theta) = \log(1-p) \qquad B(y) = 0$$

Definition of the GLM

A Generalized Linear Model for the data $(x_1, Y_1), \ldots, (x_n, Y_n)$ consists in

1) an exponential model for $Y \in \mathbb{R}$
2) a linear predictor $\eta_i = x_i^T \beta$, (with $\beta \in \mathbb{R}^p$)
3) a link function such that $g(\mathbb{E}(Y_i)) = x_i^T \beta$

Remark:

- This model allows to link Y_i with the linear predictor $\eta_i = x_i^T \beta$, assuming that $\mu_i = \mathbb{E}(Y_i) = g^{-1}(\eta_i) = g^{-1}(x_i^T \beta)$ for some link function g.
- The link between the linear predictor and the exponential model is through the expectation of Y and the link function g.
- There are many possible choices for the function g, but it must take all the values that $\mathbb{E}(Y)$ can handle. However, there is a particular choice for the link function g: the canonical link function.

The link function g is assumed to be strictly monotonic and differentiable.

The link function

Definition

Let Y be a random variable with a density function $f_\theta(y) = \exp(\theta T(y) - A(\theta) + B(y))$, then the canonical link function is defined by $\theta = g(\mu)$, where $\mu = \mathbb{E}(Y) = A'(\theta)$. If A' is invertible, $g = (A')^{-1}$.

Remark: When we choose the canonical link function, we have $\theta = g(\mu) = x^T\beta$, i.e., $\theta_i = g(\mu_i) = x_i^T\beta$.

Examples

$Y_1, \ldots, Y_n$ n independent variables with distribution f_{θ_i} belonging to the exponential family, where θ_i is the parameter of the exponential family

1) Normal distribution $\mathcal{N}(\mu_i, \sigma^2)$:

$$f_{\mu_i}(y_i) = \text{à compléter}$$

2) Bernoulli distribution $\mathcal{B}(\pi_i)$:

$$f_{\pi_i}(y_i) = \text{à compléter}$$

3) Poisson distribution $\mathcal{P}(\lambda_i)$:

$$f_{\lambda_i}(y_i) = \text{à compléter}$$

$$Y_i \sim \mathcal{N}(\mu_i, \sigma^2)$$

$$f_{\mu_i}(y_i) = \exp\left(-\frac{1}{2}\frac{\mu_i^2}{\sigma^2}\right) \exp\left(-\frac{1}{2}\frac{y_i^2}{\sigma^2} - \frac{1}{2}\ln(2\pi\sigma^2)\right)$$

$$\times \exp\left(\frac{y_i \mu_i}{\sigma^2}\right)$$

$$g(\mu_i) = \theta_i = \mathbb{E}[Y_i] = \mu_i$$

$$\Rightarrow g(x) = x$$

2) $Y_i \sim \mathcal{B}(\pi_i)$

$$f_{\pi_i}(y_i) = \pi_i^{y_i} (1-\pi_i)^{1-y_i} = (1-\pi_i) \exp\left(y_i \ln \frac{\pi_i}{1-\pi_i}\right)$$

$$\theta_i = \ln\left(\frac{\pi_i}{1-\pi_i}\right) \qquad g(x) = \ln\left(\frac{x}{1-x}\right)$$

$$\theta_i = g(\pi_i) = x_i^T \beta$$

$$x_i^T \beta = \ln \frac{\pi_i}{1-\pi_i} \iff \pi_i = \frac{e^{x_i^T \beta}}{1+e^{x_i^T \beta}}$$

3) $Y_i \sim \mathcal{P}(\lambda_i)$

$$f_{\lambda_i}(y_i) = \frac{\lambda_i^{y_i} e^{-\lambda_i}}{y_i!} = \exp(-\lambda_i) \frac{1}{y_i!} \exp(y_i \ln \lambda_i)$$

$$\theta_i = \ln \lambda_i, \quad g(x) = \ln(x)$$

Moments

$\ell_i(\theta_i; y_i) = \ln f_{\theta_i}(y_i)$: the contribution of the ith observation to the log-likelihood

$$\ell_i(\theta_i, y_i) = \theta_i T(y_i) - A(\theta_i) + B(y_i)$$

We have

$$\frac{\partial \ell_i}{\partial \theta_i} = T(y_i) - A'(\theta_i) \quad \text{and} \quad \frac{\partial^2 \ell_i}{\partial \theta_i^2} = -A''(\theta_i)$$

Under regularity assumptions (differentiation under the integral),

$$\mathbb{E}\left(\frac{\partial \ell}{\partial \theta}\right) = 0 \quad \text{and} \quad -\mathbb{E}\left(\frac{\partial^2 \ell}{\partial \theta^2}\right) = \mathbb{E}\left[\left(\frac{\partial \ell}{\partial \theta}\right)^2\right].$$

Then,

$$\mathbb{E}[T(Y_i)] = A'(\theta_i) \quad \text{and} \quad \text{Var}[T(Y_i)] = A''(\theta_i)$$

$$\mathbb{E}\left[\frac{\partial \ell}{\partial \theta}(\theta, Y)\right] = \int \frac{\partial \ln f_\theta(y)}{\partial \theta} f_\theta(y)\, dy$$

$$= \int \frac{\partial f_\theta(y)}{\partial \theta} \times \frac{1}{f_\theta(y)} f_\theta(y)\, dy$$

$$= \frac{\partial}{\partial \theta} \underbrace{\int f_\theta(y)\, dy}_{=1} = 0$$

$$\mathbb{E}\left[\frac{\partial^2 \ell}{\partial \theta^2}\right] = \mathbb{E}\left[\frac{\partial}{\partial \theta}\left(\frac{\partial \ln f_\theta(Y)}{\partial \theta}\right)\right]$$

$$= \mathbb{E}\left[\frac{\partial}{\partial \theta}\left(\frac{1}{f_\theta(y)}\frac{\partial f_\theta(y)}{\partial \theta}\right)\right]$$

$$= \mathbb{E}\left[-\frac{1}{f_\theta(y)^2}\left(\frac{\partial f_\theta(y)}{\partial \theta}\right)^2 + \frac{1}{f_\theta(y)}\frac{\partial^2 f_\theta(y)}{\partial \theta^2}\right]$$

$$= \mathbb{E}\left[-\left(\frac{\partial \ln f_\theta}{\partial \theta}\right)^2\right] + \int \frac{1}{f_\theta(y)}\frac{\partial^2 f_\theta(y)}{\partial \theta^2} \times f_\theta(y)\,dy$$

$$= \mathbb{E}\left[-\left(\frac{\partial \ln f_\theta}{\partial \theta}\right)^2\right] + \frac{\partial^2}{\partial \theta^2}\underbrace{\int f_\theta(y)\,dy}_{=1} = 0$$

$$= \mathbb{E}\left[-\left(\frac{\partial \ell}{\partial \theta}\right)^2\right]$$

Estimation

Recall that: $\eta_i = x_i^T \beta = g(\mu_i)$ where $\mu_i = \mathbb{E}(Y_i)$
The estimators of the β_j are calculated by maximizing the log-likelihood.
Log-likelihood:

$$\mathscr{L}(\beta) = \sum_{i=1}^{n} \ln f_{\theta_i}(y_i) = \sum_{i=1}^{n} \ell_i(\theta_i; y_i)$$

We have:

$$\frac{\partial \ell_i}{\partial \beta_j} = \frac{\partial \ell_i}{\partial \theta_i} \frac{\partial \theta_i}{\partial \mu_i} \frac{\partial \mu_i}{\partial \eta_i} \frac{\partial \eta_i}{\partial \beta_j}$$

Estimation

If $T(y) = y$, we have

$$\frac{\partial \ell_i}{\partial \theta_i} = y_i - A'(\theta_i) = y_i - \mu_i$$

$$\frac{\partial \mu_i}{\partial \theta_i} = A''(\theta_i) = \mathrm{Var}(Y_i)$$

$$\frac{\partial \eta_i}{\partial \beta_j} = x_{ij}$$

$\frac{\partial \mu_i}{\partial \eta_i}$ depend on the link function $\eta_i = g(\mu_i)$

Likelihood equations:

$$\sum_{i=1}^{n} \frac{(y_i - \mu_i) x_{ij}}{\mathrm{Var}(Y_i)} \frac{\partial \mu_i}{\partial \eta_i} = 0, \quad j = 1, \ldots, p$$

$\rightsquigarrow$ Non linear equations in β that need iterative methods to be solved

Particular case: the canonical link function

In that case : $\eta_i = \theta_i = x_i^T \beta$ and

$$\frac{\partial \mu_i}{\partial \eta_i} = \frac{\partial \mu_i}{\partial \theta_i} = \frac{\partial A'(\theta_i)}{\partial \theta_i} = A''(\theta_i)$$

Then,

$$\frac{\partial \ell_i}{\partial \beta_j} = \frac{y_i - \mu_i}{\mathrm{Var}(Y_i)} A''(\theta_i) x_{ij} = (y_i - \mu_i) x_{ij}$$

The equations become: $X^T y = X^T \mu$

In the gaussian case with $\mu = X\beta$ (the link function is the identity), we obtain the solution

$$\hat{\beta} = (X^T X)^{-1} X^T y$$

which corresponds to the ordinary least squares estimator

Convexity of the negative log-likelihood

For a canonical GLM with $T(y) = y$, let

$$L_n(\beta) = -\ell_n(\beta) = \frac{1}{n}\sum_{i=1}^{n}\left[A(x_i^T\beta) - y_i x_i^T\beta - B(y_i)\right].$$

Then, with $\mu_i(\beta) = A'(x_i^T\beta)$,

$$\nabla L_n(\beta) = \frac{1}{n}X^T(\mu(\beta) - y), \quad \nabla^2 L_n(\beta) = \frac{1}{n}X^T W(\beta) X,$$

where $W(\beta) = \mathrm{diag}(A''(x_i^T\beta))$.

Key observation

Since $A''(\theta) = \mathrm{Var}_\theta(Y) \geq 0$,

$$\forall v \in \mathbb{R}^p, \quad v^T\nabla^2 L_n(\beta) v = \frac{1}{n}\sum_{i=1}^{n} A''(x_i^T\beta)(x_i^T v)^2 \geq 0.$$

Thus L_n is convex.

Solving the likelihood equations

Iteratively Reweighted Least Squares (IRLS)

To compute the Maximum Likelihood Estimator (MLE), we use a Newton-Raphson type algorithm.

At iteration k, let $\hat{\beta}^k$ denote the current estimate. We approximate the normalized log-likelihood $\ell_n = \frac{1}{n}\ell$ by its second-order Taylor expansion:

$$\ell_n(\hat{\beta}^k + h) \approx \ell_n(\hat{\beta}^k) + \nabla \ell_n(\hat{\beta}^k)^T h + \frac{1}{2} h^T \nabla^2 \ell_n(\hat{\beta}^k) h.$$

Maximizing this quadratic approximation gives

$$h^* = -\left[\nabla^2 \ell_n(\hat{\beta}^k)\right]^{-1} \nabla \ell_n(\hat{\beta}^k).$$

We update

$$\hat{\beta}^{k+1} = \hat{\beta}^k + h^*$$

and repeat until convergence.

Theoretical results

We introduce the average Fisher information $I_n(\beta) = -\mathbb{E}[\nabla^2 \ell_n(\beta)]$, where ℓ_n is the average log-likelihood. Assume $I_n(\beta^*) \to I(\beta^*) \succ 0$.

Consistency and asymptotic normality

For fixed p, under appropriate regularity and design assumptions (cf. Fahrmeir and Kaufmann, 1985 [1]), for the true parameter β^*,

- $\hat{\beta} \xrightarrow{\mathbb{P}} \beta^*$
- $\hat{\beta}$ is asymptotically gaussian

$$\mathbb{P}(\beta^* \in IC(\alpha)) \geq 1-\alpha$$

$$\sqrt{n}(\hat{\beta} - \beta^*) \xrightarrow{\mathcal{L}} \mathcal{N}(0, I(\beta^*)^{-1})$$

The assumptions include identifiability and a non-degenerate limiting information matrix, bounded covariates alone do not suffice.

$\sqrt{n}\, \mathcal{I}(\beta^*)^{1/2}(\hat{\beta} - \beta^*) \to \mathcal{N}(0, I_p)$ ⇒ $\sqrt{n}\, \mathcal{I}(\hat{\beta})^{1/2}(\hat{\beta} - \beta^*) = \sqrt{n}\, \mathcal{I}(\beta^*)^{1/2}(\hat{\beta} - \beta^*) \times \mathcal{I}(\beta^*)^{-1/2}\mathcal{I}(\hat{\beta})^{1/2}$

↳ dépend de β^*

High-dimensional GLMs and the Lasso

si $\mathcal{I}(\beta^*)^{-1/2}\mathcal{I}(\hat{\beta})^{1/2} \xrightarrow{\mathbb{P}} 1$
on peut appliquer Slutsky

$$\text{et } \beta^* \in \left[\hat{\beta} \pm q_{1-\alpha/2}(\mathcal{N}(0,1))\frac{\mathcal{I}(\hat{\beta})^{-1/2}}{\sqrt{n}}\right]$$

When p is large (possibly $p > n$), the unpenalized maximum likelihood estimator may not be uniquely defined or may fail to exist as a finite vector.

The Lasso estimator

For $\lambda > 0$, consider

$$\hat{\beta}_\lambda \in \operatorname*{arg\,min}_{\beta \in \mathbb{R}^p} \left\{ L_n(\beta) + \lambda \|\beta\|_1 \right\}.$$

$$L_n(\beta) = -\ell_n(\beta) = -\frac{1}{n}\ell(\beta)$$

- The ℓ_1 penalty promotes sparsity (variable selection).
- With a canonical link, the objective is convex.
- Unlike Gaussian least squares, L_n is generally not quadratic.

Convention: any intercept is left unpenalized.

Optimality conditions for the GLM Lasso

By convexity, $\hat{\beta}_\lambda$ minimizes $L_n(\beta) + \lambda\|\beta\|_1$ if and only if

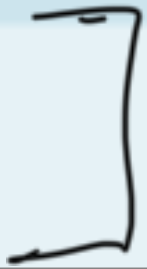
$$0 \in \nabla L_n(\hat{\beta}_\lambda) + \lambda \partial \|\hat{\beta}_\lambda\|_1.$$

For the canonical link, $\nabla L_n(\beta) = X^T(\mu(\beta) - y)/n$. Thus

$$\frac{1}{n} X^T(y - \hat{\mu}) \in \lambda \partial \|\hat{\beta}_\lambda\|_1.$$

Coordinatewise KKT conditions

$$\begin{cases} \frac{1}{n} X_j^T(y - \hat{\mu}) = \lambda \operatorname{sgn}(\hat{\beta}_{\lambda,j}), & \hat{\beta}_{\lambda,j} \neq 0, \\ \left| \frac{1}{n} X_j^T(y - \hat{\mu}) \right| \leq \lambda, & \hat{\beta}_{\lambda,j} = 0. \end{cases}$$



Question: what changes compared with linear regression?

Application : logistic regression

Model: $Y_i \mid x_i \sim \mathscr{B}(\pi_i)$, independently, with

$$\log \frac{\pi_i}{1-\pi_i} = x_i^T \beta, \qquad \pi_i = \frac{e^{x_i^T \beta}}{1 + e^{x_i^T \beta}}.$$

To derive at the board

1. Write the negative log-likelihood $L_n(\beta)$.
2. Compute its gradient and Hessian.
3. Show convexity and write the logistic Lasso estimator.
4. Derive the KKT conditions.

Board work

Le modèle logistique :

$Y_i \in \{0,1\}$, covariables $x_i \in \mathbb{R}^p$

$Y_i \sim \mathcal{B}(\pi_i)$, $\pi_i = \mathbb{P}(Y_i = 1)$

Il faut faire le lien entre π_i et x_i

- prédicteur linéaire $\eta_i = x_i^T \beta$ ($\beta \in \mathbb{R}^p$ à estimer)
- fonction de lien g tq $g(\mathbb{E}(Y_i)) = g(\pi_i) = x_i^T \beta$

  $\pi_i \in [0,1]$ et $x_i^T \beta \in \mathbb{R}$

  il faut donc une fonction $g : [0,1] \longrightarrow \mathbb{R}$

  qui soit monotone et continue

De telles fonctions sont les fonctions quantiles $g = F^{-1}$ où $F : \mathbb{R} \to [0,1]$ fonction de répartition d'une certaine loi.

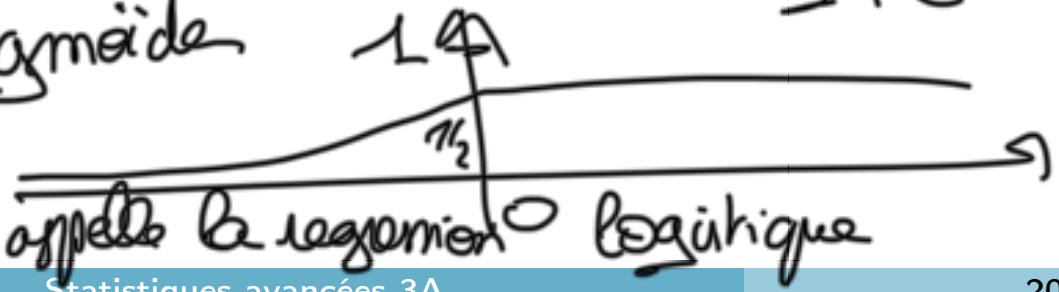
Par exemple si $F = \Phi$ fonction de répartition de $\mathcal{N}(0,1)$
$\Rightarrow$ on obtient la régression probit.

- fonction de lien canonique pour le modèle logistique

$$f_\pi(y) = \pi^y (1-\pi)^{1-y} = \exp\left(y \log\left(\frac{\pi}{1-\pi}\right) + \log(1-\pi)\right)$$

$$\mathbb{E}[Y] = \pi \quad \text{et} \quad \theta = \log \frac{\pi}{1-\pi}$$

pour obtenir un modèle canonique, la fonction de lien à choisir $g(\pi) = \log \frac{\pi}{1-\pi}$ car alors $\theta = g(\mathbb{E}(Y))$
c'est la fonction logit et $\pi = \frac{1}{1+e^{-\theta}} = g^{-1}(\theta)$
la fonction sigmoïde



Ce modèle linéaire généralisé s'appelle la régression logistique

$$\mathbb{P}(Y=y \mid X=x) = \begin{cases} \sigma(\beta^T x) & \text{si } y=1 \\ 1-\sigma(\beta^T x) & \text{si } y=0 \end{cases}$$

On observe $(x_1, Y_1), \dots, (x_n, Y_n)$ et on veut estimer $\beta \in \mathbb{R}^p$

$\Rightarrow$ EMV

Sans perte de généralité, on peut plutôt considérer $Y \in \{-1, 1\}$

$$f_{Y|X=x}(y) = \begin{cases} \sigma(\beta^T x) & \text{si } y=1 \\ 1-\sigma(\beta^T x) & \text{si } y=-1 \end{cases}$$

σ fonction de répartition d'une loi symétrique

donc $\sigma(-z) = 1-\sigma(z)$

$$f_{Y|X=x}(y) = \sigma(y \beta^T x)$$

La vraisemblance de $(Y_1, \dots, Y_n)$ avec les Y_i indépendantes est donnée par $L : \beta \mapsto \prod_{i=1}^{n} \sigma(y_i x_i^T \beta)$

et la log-vraisemblance :

$$-\ell(\beta) = -\sum_{i=1}^{n} \log \sigma(y_i x_i^T \beta)$$

$$= \sum_{i=1}^{n} \log\left(1 + e^{-y_i x_i^T \beta}\right)$$

$$\Rightarrow -\frac{1}{n}\ell(\beta) = \frac{1}{n}\sum_{i=1}^{n} \log\left(1 + e^{-y_i x_i^T \beta}\right)$$

Pour estimer β, on cherche à minimiser la fonction $-\frac{1}{n}\ell(\beta)$

Posons $g(z) = \log(1 + e^{-z})$ fonction de perte logistique

Interprétation : si $y_i x_i^T \beta \gg 0$, le prédicteur linéaire $x_i^T \beta$ est | très positif et $y_i = 1$ / très négatif et $y_i = -1$ $\Rightarrow \beta$ permet de bien prédire Y_i

si $y_i x_i^T \beta \ll 0$, le signe de $x_i^T \beta$ est fortement en désaccord avec celui de $y_i \Rightarrow \beta$ se trompe fortement sur l'échantillon i.

En minimisant $\beta \mapsto -\frac{1}{n}\ell(\beta)$, on cherche un β qui en moyenne sur l'échantillon considéré a de petites pertes mesurées par la fonction de perte logistique.

C'est un problème d'optimisation convexe :

$$\varphi'(z) = \frac{-1}{1+e^z} = -\sigma(-z)$$

$$\varphi''(z) = \frac{e^z}{(1+e^z)^2} = \sigma(z)\sigma(-z) = \sigma(z)(1-\sigma(z)) \in \left]0, \frac{1}{4}\right]$$

Donc $\nabla_\beta\left(-\frac{1}{n}\ell(\beta)\right) = \frac{1}{n}\sum_{i=1}^n \nabla_\beta \varphi(y_i x_i^T\beta) = -\frac{1}{n}\sum_{i=1}^n y_i \sigma(-y_i x_i^T\beta) x_i$

Hessienne

$$\nabla_\beta^2\left(-\frac{1}{n}\ell(\beta)\right) = \frac{1}{n}\sum_{i=1}^n \nabla_\beta^2(y_i x_i^T\beta) = \frac{1}{n}\sum_{i=1}^n \sigma(y_i x_i^T\beta)(1-\sigma(y_i x_i^T\beta)) \times x_i x_i^T$$

$x_i x_i^T \geq 0 \quad 0 < \sigma(1-\sigma) \leq 1/4 \quad 0 \leq \nabla_\beta^2\left(-\frac{1}{n}\ell(\beta)\right) \leq \frac{1}{4n}\sum x_i x_i^T$

From linear to logistic Lasso: curvature

⇒ Hessienne sera définie positive ⇒ pb convexe

Gaussian linear model (loss $\|y - X\beta\|_2^2/(2n)$):

$$\nabla^2 L_n(\beta) = \frac{1}{n} X^T X \quad \text{(constant Hessian)}.$$

Logistic regression:

$$\nabla^2 L_n(\beta) = \frac{1}{n} X^T W(\beta) X, \qquad W_{ii}(\beta) = \pi_i(\beta)(1 - \pi_i(\beta)).$$

$$0 < W_{ii}(\beta) \leq \frac{1}{4}, \quad \text{but} \inf_{\beta \in \mathbb{R}^p} W_{ii}(\beta) = 0 \quad (x_i \neq 0).$$

Consequence for oracle inequalities

Under a Restricted Eigenvalue condition and additional boundedness assumptions on the covariates and linear predictors, faster oracle inequalities can be established, with a rate of order $s\log(p)/n$ [Kwemou, 2016 [2]].

Tutorial: an oracle inequality for the logistic Lasso.

References I

$$\hat{\beta}_\lambda = \underset{\beta \in \mathbb{R}^p}{\operatorname{argmin}} \frac{1}{n} \sum_{i=1}^{n} \log\left(1 + e^{-y_i x_i^T \beta}\right) + \lambda \|\beta\|_2^2$$

$\Rightarrow$ pb n'hiérarchiquement convexe
$\Rightarrow$ unique solution



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